📘 MODULE 25: SEARCH HEAD

25.1 Concept

The **Search Head (SH)** is the Splunk instance users log in to. It **parses your SPL, sends the work to indexers, and combines the results**. It hosts the UI, users, roles, apps, and knowledge objects.

25.2 How a search is processed

```
User SPL → Search Head
   ├─ sends the search (+ knowledge bundle) to indexers ("map" phase)
   ├─ indexers read their own buckets, filter, and partially aggregate (in parallel)
   └─ SH merges and finishes the processing ("reduce" phase) → results
```

- **Knowledge bundle:** the SH replicates search-time knowledge (lookups, props, etc.) to the indexers so they can apply it.
- **Standalone lab:** one instance does both roles.

25.3 Command types (frequent interview topic)

Type	What it does	Examples
Generating	Starts a search, with no input events (leading `|`)	`inputlookup`, `tstats`, `metadata`, `rest`, `makeresults`
Streaming	Processes each event independently, and can run on indexers (fast)	`eval`, `rex`, `where`, `fields`, `lookup`
Transforming	Turns events into statistics tables, which finish on the SH	`stats`, `chart`, `timechart`, `top`
Non-streaming/dataset	Needs all data, so can't be distributed fully	`sort`, `dedup` (partly), `transaction`

Practical rule: put streaming commands early and transforming commands after them, and filter as early as possible.

25.4 Search modes

Mode	Behavior
Fast	Minimal field discovery (fastest)
Smart (default)	Adapts to the search type
Verbose	Full field discovery, and all events for transforming searches

If fields are "missing," check the mode.

25.5 Search jobs

- Each run creates a **job** with an ID. View under Activity → Jobs.
- Ad hoc search results are kept for a default period (**10 minutes**, extendable) before they expire.
- **Job Inspector:** Job → Inspect Job shows the time spent in each phase, used in performance tuning.

25.6 Users, roles, permissions (preview of Modules 32 and 35)

- **Default roles** include `admin`, `power`, `user` (plus `can_delete`; Splunk Cloud has extra roles such as `sc_admin`).
- A role defines: searchable indexes, default indexes, capabilities, search quotas.
- Users inherit capabilities from roles. You can import roles into other roles.

25.7 Knowledge object sharing

Every knowledge object has a **permission scope**:

Scope	Visible to
Private	Only the owner
App	Everyone with access to that app
Global (All apps)	Everyone in every app (needs a role with the right capability to share)

Most "works for me, not for you" issues are Private scope.

25.8 Hands-on

1. Run a search with the Job Inspector open: `index=_internal | stats count by sourcetype`. Note the phases.
2. Run `index=_internal | head 100 | table _time host sourcetype` in **Fast** vs **Verbose** mode and compare fields shown in the sidebar.
3. Classify these as generating/streaming/transforming: `tstats`, `eval`, `stats`, `inputlookup`, `rex`, `top`.
4. Settings → Roles → create a role `soc_l1` with `srchIndexesAllowed` = `linux_auth`, then create a user with that role and confirm they can't search `_internal`.

25.9 Troubleshooting

Symptom	Cause
User can't see an index	Role's allowed indexes
Knowledge object missing for others	Private scope or wrong app
Search slow/waiting	Concurrent search limits reached (per-role quotas, system limits)
Fields missing	Search mode Fast, or add-on not installed on the SH

25.10 Interview Q&A

- **What does the search head do?** Accepts searches, distributes them to indexers, merges results, hosts the UI and knowledge objects.
- **Streaming vs transforming?** Streaming works per event and can be distributed; transforming builds statistics tables.
- **What is a knowledge bundle?** Search-time config replicated from the SH to the indexers.
- **Search modes?** Fast, Smart, Verbose.
- **Private vs app vs global sharing?** Visibility scope of a knowledge object.

25.11 Revision

SH = UI + search coordination + knowledge · map on indexers, reduce on SH · generating / streaming / transforming · Fast/Smart/Verbose · Private/App/Global.